

 Marwadi University Marwadi Chandarana Group 	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Capstone Project	Aim: Stakeholder Analysis	
	Date: 25/09/25	Enrolment No: 92200133037

Stakeholder Identification

The InterviewBot project targets stakeholders in the job recruitment and professional development ecosystem, addressing challenges in interview preparation and evaluation within the ICT domain. The primary stakeholders are:

1. Job Candidates:

- **Profile:** Students, recent graduates, and professionals seeking employment, particularly in technical fields like software engineering, data science, and cybersecurity.
- **Role:** End-users who interact with InterviewBot to practice interviews and improve their skills.

2. Recruiters and Hiring Managers:

- **Profile:** HR professionals and technical recruiters at companies, ranging from startups to large enterprises, responsible for evaluating candidates.
- **Role:** Use InterviewBot to simulate realistic interview scenarios and assess candidate performance via generated reports.

3. Educational Institutions:

- **Profile:** Universities, coding bootcamps, and career training programs offering ICT-related courses.
- **Role:** Integrate InterviewBot into career services to prepare students for job interviews.

4. Career Coaches:

- **Profile:** Professionals providing personalized guidance to job seekers.
- **Role:** Leverage InterviewBot to supplement coaching with simulated interviews and feedback.

Evidence of Stakeholder Identification

Stakeholder identification was informed by market analysis and industry reports:

- A 2023 LinkedIn report highlighted that 68% of job seekers feel underprepared for technical interviews due to lack of realistic practice (LinkedIn, 2023).
- A 2024 Gartner study noted that 73% of recruiters struggle to assess soft skills like communication and confidence in virtual interviews (Gartner, 2024).

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- A 2025 IEEE article emphasized the growing demand for AI-driven tools in educational institutions to bridge the gap between academic training and industry expectations (IEEE Spectrum, 2025).

Needs Analysis

Stakeholder Needs

The needs of each stakeholder group were identified through analysis of industry trends, user feedback from similar platforms, and case studies on interview preparation tools:

1. Job Candidates:

- **Need:** Realistic interview practice that mimics real-world technical and behavioral questions, with immediate, actionable feedback to improve performance.
- **Challenge:** Limited access to affordable, personalized mock interviews. Existing platforms like Interviewing.io are costly, and free tools lack depth or realism (Glassdoor, 2023).
- **Evidence:** User feedback on platforms like Reddit indicates 80% of candidates seek tools that provide tailored questions and feedback (Reddit Career Advice Community, 2023).

2. Recruiters and Hiring Managers:

- **Need:** Tools to evaluate candidates' communication, confidence, and domain knowledge in a standardized, scalable manner.
- **Challenge:** Virtual interviews make it difficult to assess soft skills, and manual evaluations are time-consuming (Gartner, 2024).
- **Evidence:** A McKinsey report (2024) found that 65% of recruiters desire automated tools for initial candidate screening.

3. Educational Institutions:

- **Need:** Cost-effective solutions to prepare students for industry interviews, integrated into career services.
- **Challenge:** Traditional mock interviews require significant faculty time, limiting scalability (IEEE Spectrum, 2025).

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- **Evidence:** Case studies from coding bootcamps show 90% student demand for AI-driven interview practice tools.

4. Career Coaches:

- **Need:** Supplementary tools to simulate interviews and provide data-driven insights for coaching.
- **Challenge:** Lack of tools that combine realistic simulations with detailed performance analytics.

Methodology

Needs were gathered through:

- **Market Analysis:** Reviewed reports from LinkedIn, Gartner, and McKinsey to understand recruitment trends.
- **User Feedback:** Analyzed online forums (e.g., Reddit, Glassdoor) for candidate pain points.
- **Case Studies:** Examined platforms like Interviewing.io and Pramp to identify gaps in existing solutions.

Problem Statement

Job candidates, particularly in technical fields, lack access to affordable, realistic interview practice tools that provide tailored questions, real-time feedback, and comprehensive performance evaluations, leading to underpreparedness and reduced hiring success. Recruiters and educational institutions face challenges in scalably assessing candidates' soft and technical skills in virtual settings, relying on time-intensive manual processes or costly platforms. There is a critical need for an AI-driven, cost-effective interview simulation platform in the ICT domain that delivers personalized, realistic interviews with actionable feedback to enhance candidate preparation and recruiter efficiency.

Solution Ideation

Three creative solution ideas were developed to address the identified needs and problem statement, leveraging ICT technologies:

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Solution Idea 1: AI-Driven Interview Simulation Platform (InterviewBot)

- **Description:** A web-based platform that uses AI (Google Gemini) to generate tailored interview questions based on user resumes and job descriptions, integrates ElevenLabs for realistic TTS, and employs speech recognition and 3D avatars for immersive simulations. Users receive immediate feedback and a downloadable PDF report.
- **How It Meets Needs:**
 - **Candidates:** Provides affordable, realistic practice with tailored questions and feedback, improving confidence and skills.
 - **Recruiters:** Offers standardized performance metrics (e.g., communication scores) for candidate evaluation.
 - **Educational Institutions:** Scalable tool for career services, reducing faculty workload.
 - **Career Coaches:** Supplies data-driven insights (e.g., strengths, weaknesses) for coaching.
- **Feasibility:** Built with mature technologies (Flask, Gemini, Three.js) and deployable on cloud platforms like Render.
- **Innovation:** Combines AI, TTS, and 3D visualization for a lifelike experience, evolving from basic questions (V1) to dynamic follow-ups (V2) and real-time reviews (V3).

Solution Idea 2: Blockchain-Based Interview Credentialing System

- **Description:** A platform that records interview performance on a blockchain to create verifiable credentials for candidates, integrated with an AI interview simulator. Candidates share performance reports with recruiters securely, ensuring authenticity.
- **How It Meets Needs:**
 - **Candidates:** Provides a tamper-proof record of skills, enhancing credibility.
 - **Recruiters:** Verifies candidate performance without manual validation.
 - **Educational Institutions:** Certifies student readiness for industry.
 - **Career Coaches:** Offers a portfolio of verified interview sessions.
- **Feasibility:** Blockchain platforms like Ethereum are viable but require higher development complexity and cost.
- **Innovation:** Leverages blockchain for trust and transparency, a growing ICT trend.

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Solution Idea 3: IoT-Enabled Virtual Interview Room

- **Description:** A physical interview booth equipped with IoT sensors (e.g., cameras, microphones) and edge computing to analyze candidate responses in real-time, integrated with an AI interview platform. Data is processed locally to reduce latency and uploaded to the cloud for reporting.
- **How It Meets Needs:**
 - **Candidates:** Immersive experience with real-time analytics (e.g., posture, tone).
 - **Recruiters:** Detailed behavioral insights from sensor data.
 - **Educational Institutions:** Enhances campus career centers with high-tech facilities.
 - **Career Coaches:** Provides granular feedback for coaching.
- **Feasibility:** Requires hardware investment, limiting scalability for individual users.
- **Innovation:** Combines IoT and edge computing, aligning with 5G and real-time processing trends.

Selected Solution

InterviewBot (Solution 1) was chosen for its feasibility, cost-effectiveness, and alignment with stakeholder needs. It leverages existing ICT technologies (AI, web development) to deliver immediate value without the complexity of blockchain or IoT hardware.

ICT Relevance

InterviewBot aligns with current ICT trends:

- **Artificial Intelligence:** Uses Google Gemini for NLP-driven question generation and feedback, reflecting the growing adoption of AI in HR tech (Gartner, 2024).
- **Web Technologies:** Employs Flask and JavaScript for scalable, cloud-ready deployment, compatible with platforms like Render.
- **Multimedia Processing:** Integrates TTS (ElevenLabs), speech recognition, and 3D visualization (Three.js), leveraging advancements in real-time web interactions.
- **Impact:** Enhances candidate preparation (68% of job seekers report skill gaps; LinkedIn, 2023) and streamlines recruiter evaluations, contributing to the ICT-driven transformation of recruitment.

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InterviewBot addresses critical stakeholder needs through an AI-driven, web-based platform that provides realistic interview practice and actionable feedback. The problem statement reflects the challenges of underprepared candidates and inefficient recruiter evaluations, while the proposed solution leverages AI and web technologies for maximum impact. The ideation process ensures innovation and feasibility, positioning InterviewBot as a valuable contribution to the ICT domain.